Name: Date:

Standards: MA.5.AR.1.1; MA.5.AR.3.1-3.2

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 14: rate.

1 Recall: Rate vs. Ratio

Explain the difference between a ratio and a rate. Give one example of each.

Write 2 to 3 sentences.

A ratio compares two quantities of the same kind (e.g., 3 red to 5 blue marbles). A rate compares two quantities with different

2 Recall: Unit Rate

A grocery store sells 4 cans of tomatoes for $5.20.

Find the unit rate (price per can).

$5.20 ÷ 4 = $1.30 per can.

3 Application: Distance

A truck driver travels at 65 mph for 4 hours.

How far does the truck travel?

65 × 4 = 260 miles.

4 Application: Time

A swimmer needs to swim 1.5 km. She swims at 0.5 km per hour.

How many hours will it take?

1.5 ÷ 0.5 = 3 hours.

5 Word Problem: Better Buy

A 16-oz bag of pretzels costs $3.20. A 24-oz bag costs $4.56.

Find the cost per ounce for each. Which is the better deal?

16 oz: $0.20/oz. 24 oz: $0.19/oz. The 24-oz bag is the better deal.

6 Word Problem: Two-Part Trip

Kai drives 75 km/h for 2 hours, then 90 km/h for 1 hour.

What is the total distance? What is the average speed for the whole 3-hour trip?

Part 1: 150 km. Part 2: 90 km. Total = 240 km. Average speed = 240 ÷ 3 = 80 km/h.

7 Word Problem: Production

A loom weaves 8 meters of fabric per hour. A factory runs 3 looms for 5 hours.

How many total meters of fabric are produced?

1 loom × 5 h = 40 m. 3 looms = 3 × 40 = 120 m.

8 Find the Mistake

A student says: 'A car travels 150 miles in 3 hours, so its speed is 450 mph (150 × 3).'

Find the error and calculate the correct speed.

The student multiplied instead of dividing. Speed = 150 ÷ 3 = 50 mph.

9 Multi-Step: Combining Workers

Sophia wraps 20 gifts per hour. Atlas wraps 15 gifts per hour. They need to wrap 140 gifts.

Working together, how long will it take?

Combined rate = 35 gifts/hour. 140 ÷ 35 = 4 hours.

10 Reasoning: Head Start

Mira starts a 60-km bike ride at 8:00 AM going 15 km/h. Leo starts at 9:00 AM going 20 km/h on the same route.

How far ahead is Mira when Leo starts? At what time does Leo catch up?

Head start = 15 km. Closing rate = 5 km/h. Time to catch up = 15 ÷ 5 = 3 hours after 9:00 AM = 12:00 PM (noon).